MindMatrix

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Front-End UI Development Using Compose

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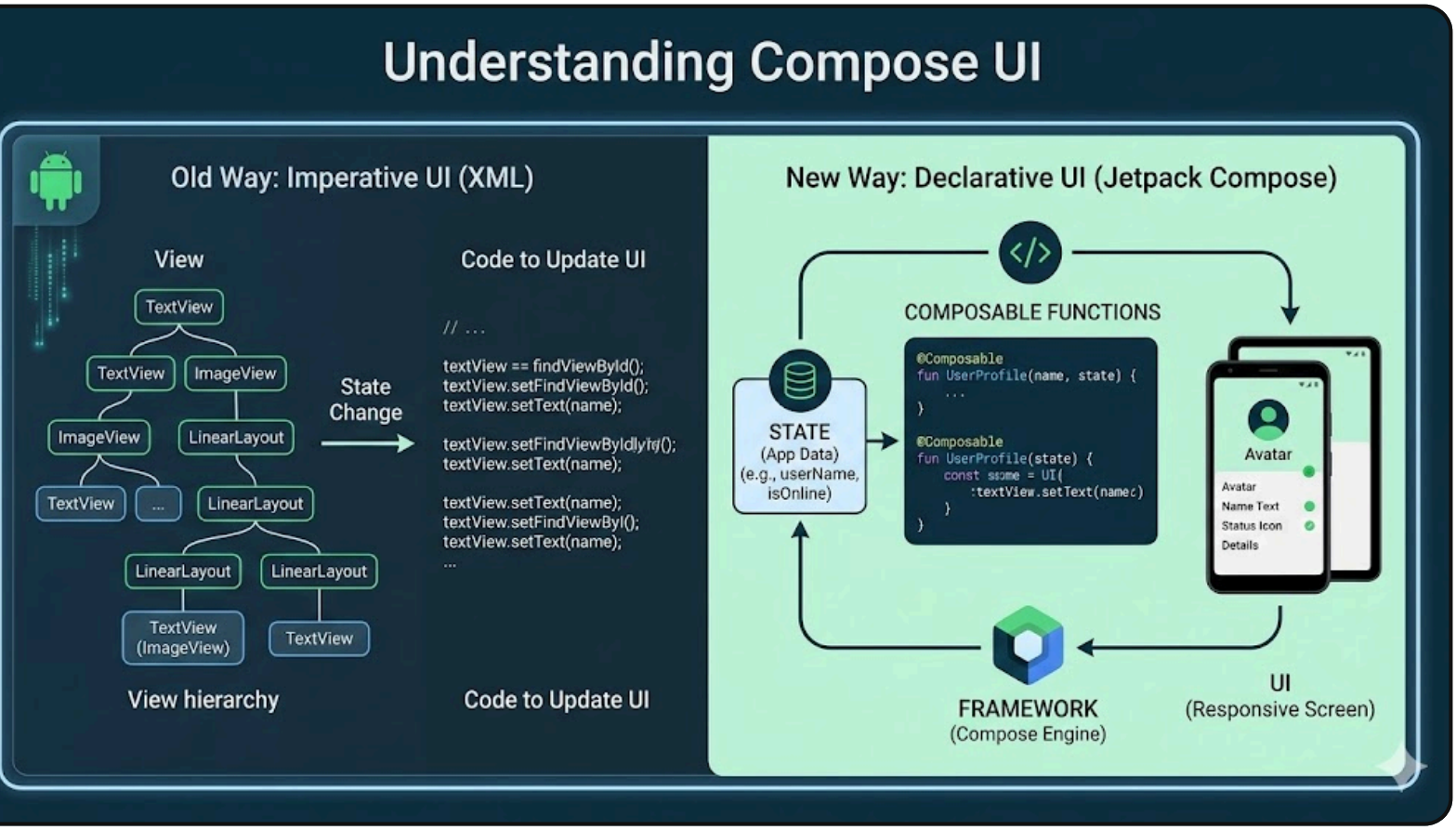


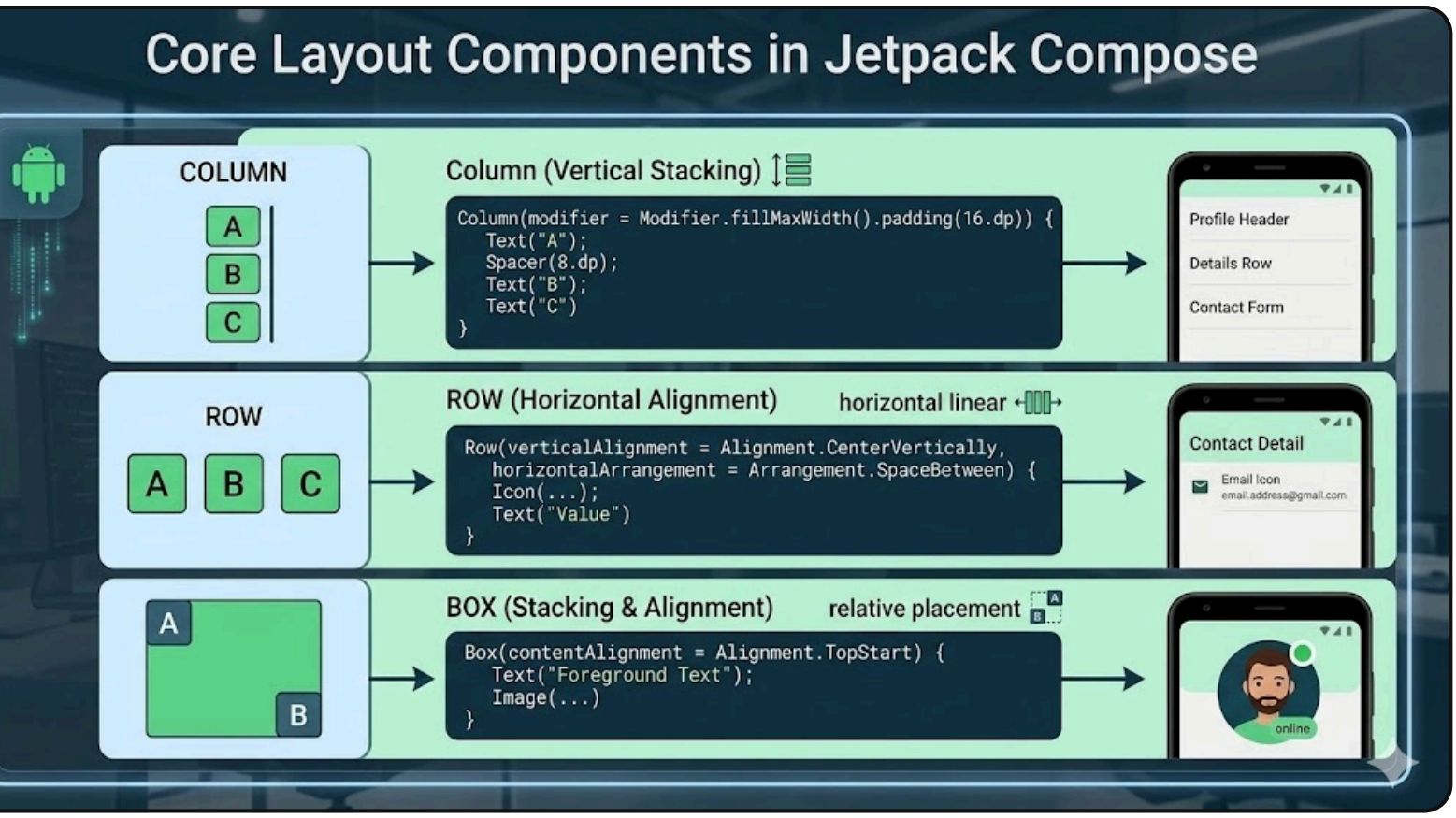
Understanding Compose UI

Jetpack Compose represents a paradigm shift in Android development, moving from an imperative model (XML) to a declarative approach. In this project, I built the UI using Composable functions (annotated with @Composable). Instead of manually manipulating view hierarchies, these functions describe the UI's state, and the framework efficiently handles the rendering and updates. This significantly reduces boilerplate code and improves maintainability.

Core Layout Components

The Business Card application relies on fundamental layout building blocks to achieve a professional look:

- Column: Used to stack the Profile and Contact sections vertically. I utilized verticalArrangement = Arrangement.SpaceBetween to ensure the contact info remained at the bottom while the profile stayed centered.
- Row: Implemented within the contact section to align icons (Phone, Share, Email) horizontally with their respective text labels.
- Modifiers: These were crucial for styling. I applied fillMaxSize() to the main container, added specific padding() for whitespace, and used Alignment.CenterHorizontally to ensure visual balance.



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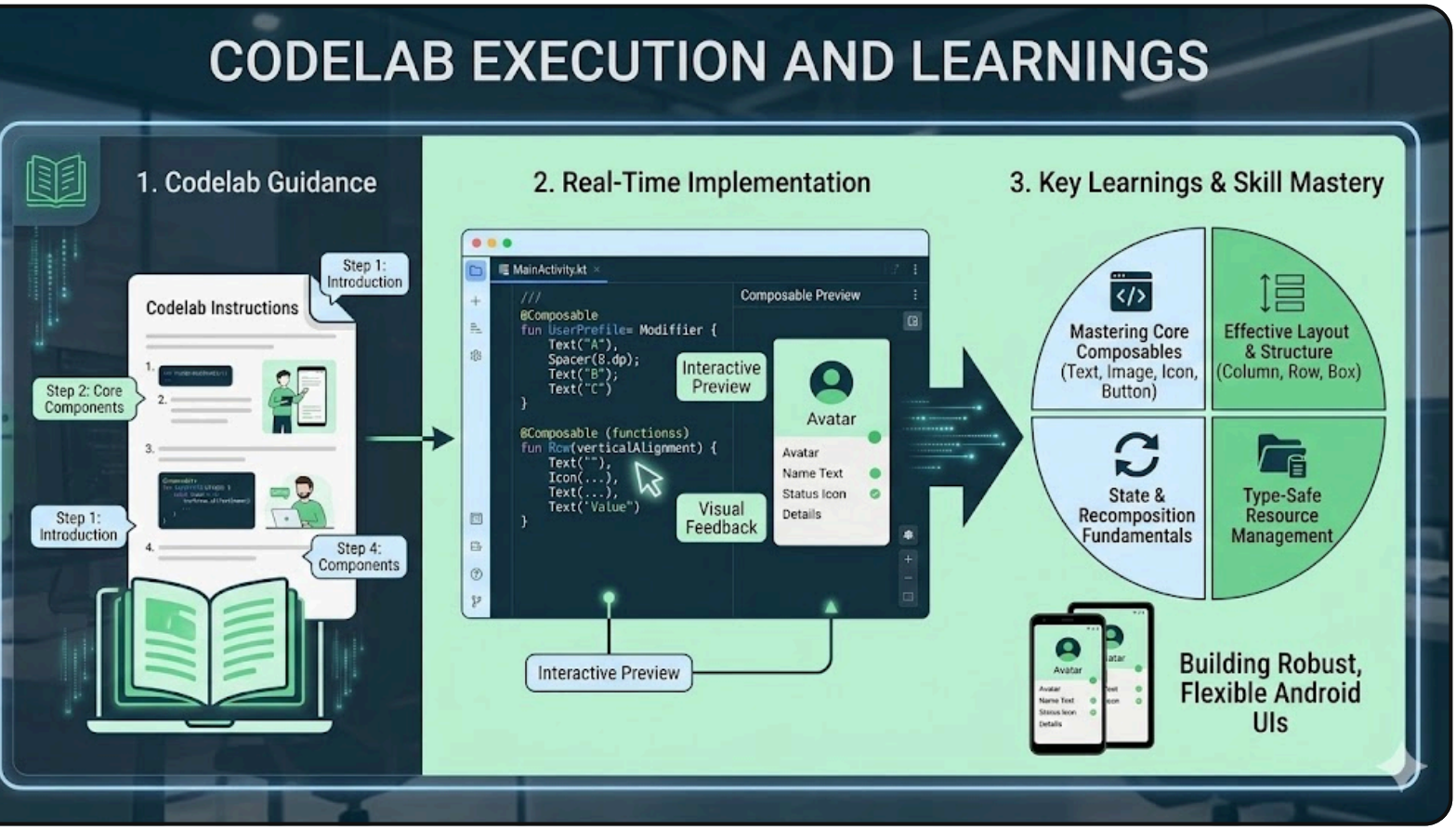


Codelab Execution and Learnings

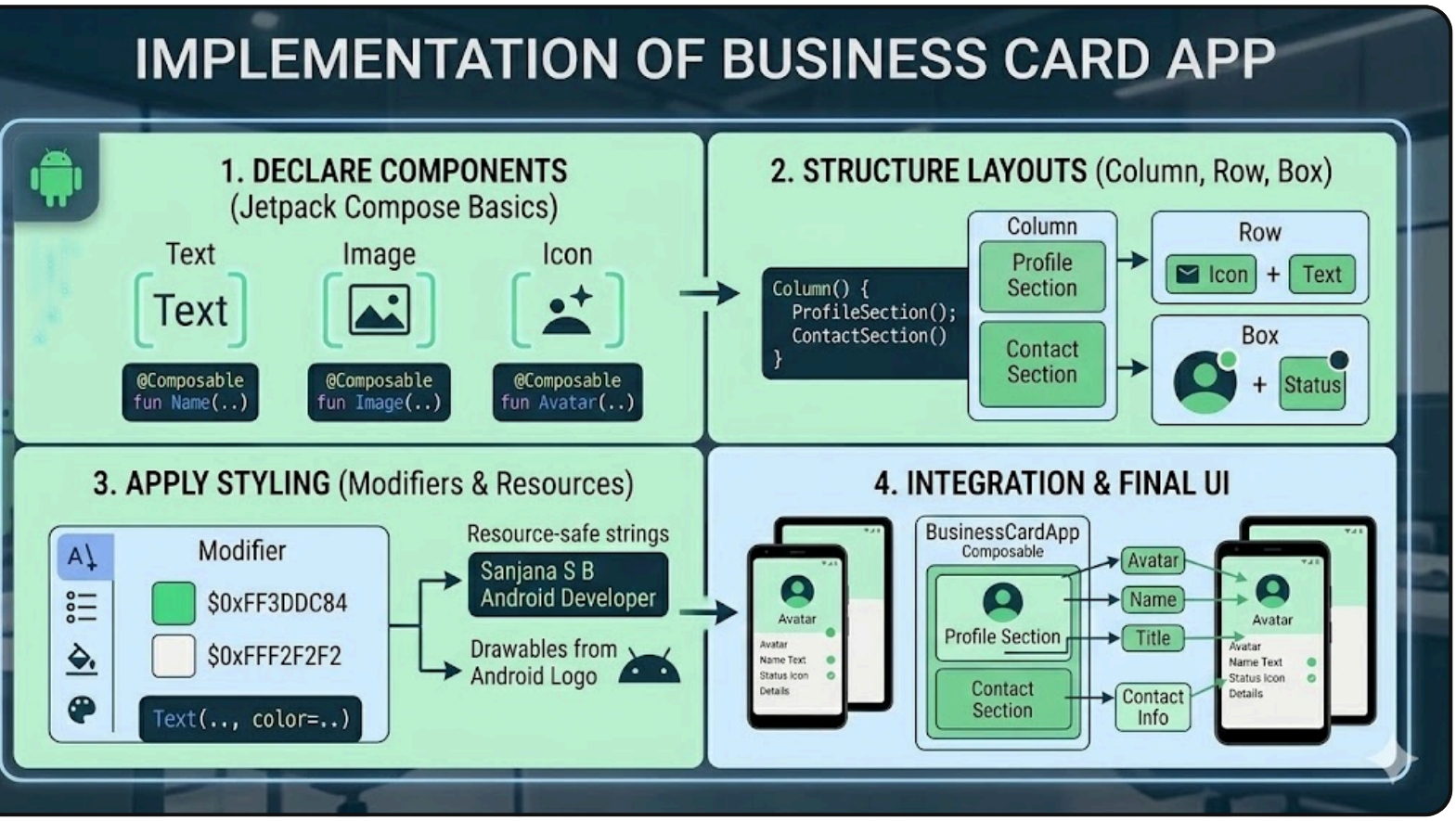
Through the completed codelabs, I gained hands-on experience with:

- Basic Composables: Implementing Text, Image, and Icon components.
- Resource Management: Accessing strings and drawables in a type-safe manner.
- The Preview Tool: Using the @Preview annotation to visualize UI changes in real-time within Android Studio, which drastically accelerated the design iteration process compared to traditional emulator-only testing.

Implementation of Business Card App

In MainActivity.kt, the BusinessCardApp function serves as the primary UI entry point. It wraps the entire interface in a Surface with a custom background color ($0xFFF2F2F2$).

- ProfileSection: Centrally displays the Android logo, name, and professional title.
- ContactSection: Uses a modular approach to display contact details. Each detail is a Row containing a tinted Icon ($0xFF3DDC84$) and a Text element with specific font sizing ($16.sp$).



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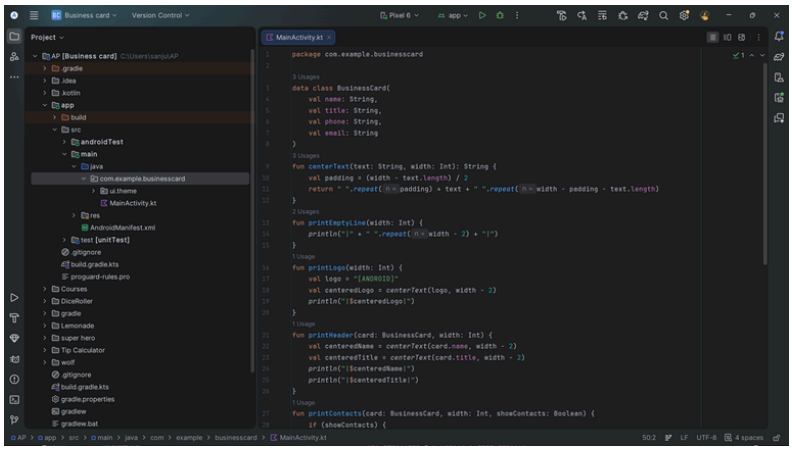
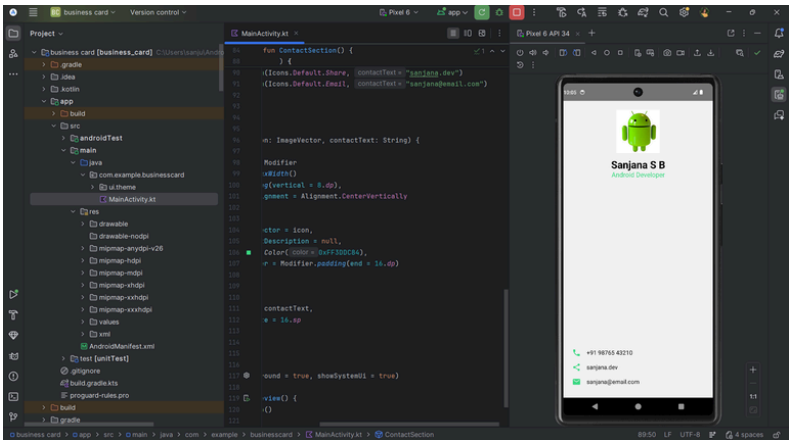
Application Screenshots

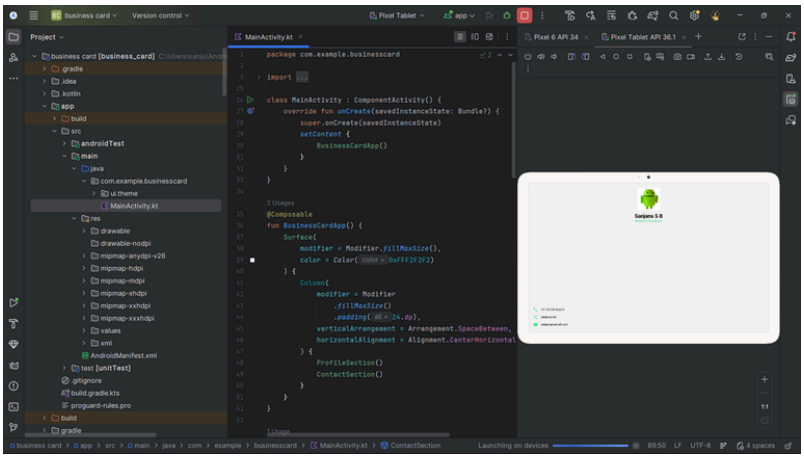
Compose UI Code



Business Card UI on Phone Emulator



Business Card UI on Tablet Emulator



Conclusion

This project demonstrated the efficiency of Jetpack Compose in building modern, responsive Android interfaces. By using a declarative syntax and powerful layout components like Columns and Rows, I was able to create a functional Business Card app that scales across different device form factors, from phones to tablets, with minimal code complexity.